

IoT (Internet of Things): IoT refers to a network of physical devices, vehicles, appliances, sensors, and other objects embedded with software, sensors, and network connectivity that allow them to collect and exchange data. IoT systems are designed to enable automation and intelligent decision-making by connecting devices to the internet, thereby creating a smart ecosystem. The key characteristics of IoT are:

- **Interconnectivity:** IoT devices are interconnected, enabling them to exchange data with one another. This could be as simple as a smart thermostat communicating with a home's heating and cooling system, or as complex as industrial machines sharing data with cloud-based analytics platforms.
- **Automation:** IoT allows for real-time decision-making and automation. For example, IoT-enabled smart homes can adjust lighting and heating based on user preferences or environmental factors.
- **Data generation:** IoT devices generate vast amounts of data, which can be used to optimise processes, improve efficiency, and enhance user experiences. In industries like healthcare or agriculture, IoT devices can track vital signs, soil moisture levels, or crop health, allowing for real-time monitoring and intervention.

However, as IoT networks grow, they face several challenges, including security vulnerabilities, lack of control over data, and dependency on centralised cloud platforms. This is where decentralisation and blockchain technology can come into play to solve these problems.

Blockchain: Blockchain is a distributed ledger technology (DLT) that records transactions in a secure, transparent, and immutable manner. Each block in a blockchain contains a set of transactions, and these blocks are linked together in a chronological order to form a chain. Blockchain's key feature is its decentralisation — there's no central authority controlling the data, and every participant (node) in the network has access to the same information. Here are the key characteristics of blockchain.

- **Immutability:** Once a transaction is recorded in a blockchain, it cannot be altered or deleted. This ensures data integrity, making it tamper-resistant.
- **Transparency:** Blockchain is public and transparent, allowing anyone to view the transactions recorded on the ledger, which is vital for trust and accountability in decentralised networks.
- **Security:** Blockchain uses cryptography to ensure that data transactions are secure and only authorised participants can add new transactions to the ledger. This makes blockchain an ideal solution for securing IoT networks, where data privacy and authenticity are critical.
- **Decentralisation:** Rather than relying on a central server, blockchain is maintained by a distributed network of

nodes. Each node validates transactions, ensuring that the system is decentralised and fault-tolerant.

When applied to IoT, blockchain ensures that data from devices is securely recorded and can be verified without relying on a central authority, which enhances security and trust among devices in the network.

Decentralisation: Decentralisation refers to distributing control across a network, removing the need for a central governing entity or authority. In traditional IoT systems, devices and data are typically managed through centralised cloud platforms or servers, creating potential single points of failure, bottlenecks, and security risks.

In a decentralised IoT network, no central server or entity controls the data flow or manages the devices. Instead, devices communicate directly with each other using decentralised protocols, and transactions or data exchanges are verified and recorded using blockchain technology. The key characteristics of decentralisation are:

- **Peer-to-peer communication:** In decentralised IoT, devices can communicate with each other directly without the need for an intermediary. This reduces latency, speeds up decision-making, and lowers the dependency on central servers.
- **Distributed control:** Instead of a single centralised authority managing devices and data, control is distributed among network participants. Each device can act as both a producer and consumer of data, ensuring that decisions are made locally, improving speed and responsiveness.
- **Resilience and redundancy:** By distributing data and control across multiple devices or nodes, decentralised IoT systems are more resilient to failures. Even if one node or device goes offline, the network can continue to function without significant disruption.
- **Privacy and security:** Decentralisation enhances privacy by allowing IoT devices to exchange data without relying on a third-party cloud service. Blockchain further strengthens privacy by encrypting and anonymising data exchanges.

Decentralised IoT versus centralised IoT

In centralised IoT systems, data is routed through a central cloud platform or server, where it is processed, stored, and analysed. While this centralised approach allows for easier management and control, it introduces several vulnerabilities.

Single points of failure: A centralised IoT system depends on a single server or cloud infrastructure, making it a target for cyberattacks. If the central server goes down or is compromised, the entire system can fail.

Data privacy concerns: In centralised systems, all data is stored in one place, raising concerns about how that data is used, shared, and protected.